

Qi Fan (Eric) Yan

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EDUCATION

University of Waterloo

Master of Mathematics in Computer Science

Sep 2024 - Aug 2026 (Expected)

Waterloo, ON Canada

- **GPA:** 94.75/100 (4.0/4.0)
- **Masters Thesis Advisor:** Raouf Boutaba
- **Coursework:** Programmable Networks, Model Serving Systems, Privacy for Data Analysis & ML, Multi-Core Prog.
- **Awards:** Canada Graduate Scholarships Master's Program, President's Graduate Scholarship

University of British Columbia (UBC)

Bachelor of Science (Honour) in Computer Science + Master of Management Dual Degree

Sep 2018 - May 2024

Vancouver, BC Canada

- **GPA:** 91.5/100 (3.92/4.0) **Major GPA:** 94.9/100 (4.0/4.0)
- **Honours Thesis Advisor:** Ivan Beschastnikh
- **Coursework:** Distributed Systems, Internet Computing, Machine Learning & Data Mining, Natural Language Processing, Advanced Software Engineering, Operating Systems & Computer Hardware, Compiler Construction
- **Awards:** Trek Scholarship (**Top 5%**), NSERC Undergraduate Student Research Award, Dean's List, Science Scholar

RESEARCH INTEREST

Distributed systems, systems/networks for ML, ML for systems/networks, federated learning

PUBLICATION

- [1] **Qifan Yan**, Andrew Liu, Shiqi He, Mathias Lécuyer, Ivan Beschastnikh. **FedFetch**: Faster Federated Learning with Adaptive Downstream Prefetching. *IEEE International Conference on Computer Communications (INFOCOM 2025)*. <https://arxiv.org/abs/2504.15366>
- [2] Shiqi He, **Qifan Yan**, Feijie Wu, Lanjun Wang, Mathias Lécuyer, Ivan Beschastnikh. **GlueFL**: Reconciling Client Sampling and Model Masking for Bandwidth Efficient Federated Learning. *Conference on Machine Learning and Systems (MLSys 2023)*. <https://arxiv.org/abs/2212.01523>

RESEARCH EXPERIENCE

Networks Lab, University of Waterloo

Sep 2024 - Present

Project: Accelerating ML With Dynamic In-Network Aggregation Trees (Ongoing)

Supervisor: Raouf Boutaba

- Designed an in-network aggregation system for ML in multi-tenant datacenter with programmable switches
- Revealed key inefficiencies in both top-of-rack switch only and per-job routing in-network aggregation systems where the former cannot use all resources in the network and the latter is slow to adapt to real-time congestion
- Developed a flowlet-based random routing protocol to construct aggregation trees in milliseconds, leveraging network-wide spare resources while introducing packet priority mechanisms for optimized memory efficiency
- Created a large-scale packet-level trace-based simulation in C++ with ns-3 for evaluating the proposed techniques effectiveness over prior solutions accompanied by a P4 prototype (currently under evaluation)

Project: Federated Reinforcement Learning for Service Continuity (Ongoing)

Supervisor: Raouf Boutaba

- Addressed service continuity challenges during multi-domain user equipment handovers within O-RAN-based network slices, specifically targeting the latency limitations of legacy models and mechanisms
- Proposed a Federated Multi-Agent Reinforcement Learning framework with client clustering to enable rapid and proactive resource allocation for Virtual Network Functions with slice/traffic-specific personalized models

Systopia Lab, University of British Columbia

Apr 2022 - Jul 2024

Project: Faster Federated Learning with Adaptive Downstream Prefetching

Supervisor: Ivan Beschastnikh

- Created **FedFetch** ([open-sourced link](#)), a scheduling algorithm for cross-device federated learning, which leverages pre-sampling of clients and pre-fetching of model updates to accelerate training via communication cost reduction
- Demonstrated a 26% end-to-end training time reduction and broad compatibility with compression algorithms by validating FedFetch in a custom simulation system based on the FedScale platform with Python and PyTorch

- Optimized pre-sampling + pre-fetching algorithms to address complications arising from client and network heterogeneity such as different bandwidth/compute capacity and online availability for federated learning

Project: Reconciling Client Sampling and Sparsification in Federated Learning

Supervisor: Ivan Beschastnikh

- Contributed to the ideation, realization, and evaluation of **GlueFL** ([open-sourced link](#)), a framework combining sticky sampling and mask shifting to reduce communication time/bandwidth costs by 27% in cross-device FL
- Improved robustness of the GlueFL framework to system heterogeneity in FL by extracting network environment data from sources including Measurement Lab, FedScale and various paper artifacts and adapting the GlueFL framework to account for the observations
- Oversaw all hyper-parameter experiments, diagnosed system and statistical performance issues, created various visualization of experiment results, and discussed results in the final paper submission

TEACHING EXPERIENCE

University of Waterloo | Teaching Assistant

Sep 2024 - Present

- CS 350 Operating Systems, CS 346 Application Development, CS 230 Introduction to Computers and Computer Systems, CS 246 Object Oriented Software Development

University of British Columbia | Teaching Assistant

Jan 2020 - Apr 2023

- CPSC 416 Distributed Systems, CPSC 317 Internet Computing, CPSC 311 Definition of Programming Languages, CPSC 210 Software Construction

INDUSTRY EXPERIENCE

Amazon | Software Development Engineer Intern

Jun 2022 – Aug 2022

- Implemented new back-end systems with Java and the Spring framework and related front-end web pages with JavaScript and Angular to enable new payment disbursement options for 20% of Amazon.com retail sellers
- Created and presented a detailed design document that went through multiple rounds of review to critically evaluate multiple design choices for the new system based on extensibility, maintainability and scalability
- Leveraged internal tools and AWS technologies including S3, Lambda, CloudWatch, and EC2 in the design, implementation and evaluation phases to verify system correctness and performance metrics

Deloitte | Business Analyst

Jan 2022 – Apr 2022

- Assisted project proposal development for migrating a major Canadian energy firm's services to cloud platforms
- Developed various components of a Canadian federal class action claim system on Azure for indigenous victims

SAP | Software Developer Intern

Sep 2020 – Apr 2021

- Committed to the full-stack development of SAP Concur's next-gen homepage used by 48,000 businesses
- Researched substantial current and legacy documentation/code to build and document high-performance endpoints for the homepage's back-end services in Go as part of SAP Concur's transition to micro-services
- Implemented over a third of the front-end components and related API connection, user-permission, error handling, logging, and analytics logic using technologies like TypeScript, React and Redux
- Spearheaded the design and creation of new E2E test automation with TestCafe covering 100% of current expense features and contributed to existing internal documentation to aid future knowledge transfers

AWARDS

- **Canada Graduate Scholarships Master's Program** | \$27,000 **2025**
- **President's Graduate Scholarship** | \$10,000 **2025**
- **INFOCOM 2025 Student Travel Grant** | \$1,450 **2025**
- **MM Accelerated Career Scholarship** | \$5,000 **2023**
- **NSERC Undergraduate Student Research Award** | \$6,000 **2020**
- **UBC Computer Science Undergraduate Service Award** **2022**

TECHNICAL SKILLS

Programming languages: Python, Java, Go, C, C++, TypeScript, JavaScript, Rust, Racket, P4

Frameworks & Libraries: ns-3, PyTorch, Scikit-learn, NumPy, Pandas, gRPC, Spark, Spring, React, Angular, Redux

Technology: Linux, Git, AWS, Kubernetes, Docker, LLVM, PostgreSQL, MongoDB, HTML, CSS, Cypress, Postman, $\LaTeX$